Facilitating Electrochemical Overall Water Splitting Through Tailoring Halloysite Nanoclay-Based Nanoarchitectonics of Ionic Environment and Gas-Bubble Wettability

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Unwanted gas bubble adhesion at the electrode surface and its unfavorable microenvironment compromise water splitting performance by perturbing the access of electrolytes and electron transfer processes. While reported bubble-repellent coatings promote an early detachment of gas bubbles—they mostly fail to provide a favorable ionic microenvironment for overall electrochemical water-splitting. Here, it has experimentally and theoretically validated the impact of the synergistic association of bubble repellence and selective ionic environments on water splitting performance. Halloysite nanoclay-based coating with tailored ionic environment and gas-bubble wettability on nickel foam (NF) ensured a superior performance for hydrogen and oxygen evolution reactions (HER and OER) in alkaline medium (1 M KOH). Superaerophobic coatings on NF with cationic modifiers boost HER by polarizing water around the electrode surface, and anionic modifiers improve OER via facilitating OH^- availability, leading to 42% and 30% lower overpotentials at a current density of 100 mA cm^{-2}, respectively. Eventually, it improves gas production by >60% compared to bare NF in alkaline conditions at a given potential. This strategy lowers the cell voltage required for overall water splitting by 500 ± 6 mV while sustaining a current density of 100 mA cm^{-2}.

1. Introduction

The electrochemical overall water-splitting process provides a renewable and environment-friendly energy source as it does not leave any carbon footprint.[1–3] However, an elevated overpotential for hydrogen and oxygen evolution reactions (HER and OER) generally demands a high operating voltage for overall water-splitting, posing a significant challenge to its widespread practical applications.[4–6] In general, electrochemical water splitting follows a heterogeneous inner-sphere electron transfer mechanism.[7–9] Hence, the HER and OER reaction kinetics are highly sensitive to the local reaction microenvironments at the electrode-electrolyte interface, where charge and electron transfer take place.[10–12] For instance, Pt nanowires displayed an anomalously high HER activity in alkaline media, attributed to the formation of localized acidic microenvironments at the catalyst surface.[13] Additionally, alkali metal cations in the electrolyte promote hydroxyl species' adsorption on platinum, affecting the HER kinetics.[14] Even, conjugated polymeric coatings decorated with hydrophilic moieties provide either an alkali-metal ion-rich environment near the electrode surface or hydrophilic non-conjugated segments provide hydrophilic pores to promote HER.[17] While these seminal reports independently validate the role of microenvironments on gas-evolving reaction performances,[18–21] they cannot associate different ionic environments for a comprehensive understanding of their implications on HER and OER performance.

On the other side, gas bubble wettability and adhesion on the electrode surface influence the charge and electron transfer processes during electrochemical gas evolution reactions (GER). In general, high gas bubble wettability and adhesion result in a delayed detachment of gas bubbles, which temporarily and repetitively obstruct the access of electrolytes to the electrochemically active sites, thereby elevating overpotentials for GER.[22–26] In this relevance, a few metal-free coatings derived from viral hydrogel deposited on Pt electrode,[22] covalently cross-linked hydrogel,[23] self-healing hydrogel[24] and polymeric nanoparticles[25] deposited on nickel foam (NF) for associating extreme bubble repellence and (or) low force of bubble adhesion. But, such earlier reports mainly reduce the overpotential for GER through the early departure of gas bubbles. Hence, further

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design of a metal-free and universal coating is essential to modulate both bubble wettability and ionic environments orthogonally, for studying their role on GER independently and unprecedentedly in the literature.

Here, we aimed to develop a metal-free and bifunctional coating on NF for facilitating electrochemical gas evolution reactions in alkaline condition (1 M KOH) through tailoring the extent of hydration, size of departing gas bubbles and ionic/non-ionic microenvironment at the electrode surface. In this relevance, a naturally available hollow nanomaterial, i.e., halloysite nanoclay (HNC), was chemically modified prior to its dip coating on the NF for associating nanoarchitectonics of ionic environment and gas-bubble wettability. The resulting porous coating retained residual chemical reactivity to modify with a range of small molecules having different hydrophilic (ionic/non-ionic) and hydrophobic (various hydrocarbon tails) moieties through a 1,4-conjugate addition reaction as illustrated in **Scheme 1A,B**. While, a bare NF suffers from high wetting and adhesion of beaded oil droplets and gas bubbles, the coated NF having appropriate post-covalent modifications with selected modifiers, provides extreme repellence to both gas bubbles and oil droplets, as depicted in Scheme 1C. It provides a facile basis to precisely modulate surface free energy (γ_{SG}) and integrate the required bubble wettability with ionic/non-ionic/hydrophobic environments. The elevation in γ_{SG} from 34.6 ± 2.1 to 52.8 ± 2.1 mN m^{-1} resulted in superaerophobicity with lowered bubble adhesion force from 158.2 ± 2.9 to 4.7 ± 0.7 μN by improving their hydration, and thereby reducing the departing gas bubbles' size (Scheme 1D). Moreover, its inherent ability of extreme oil repellence maintains nearly unperturbed alkaline water splitting performance for electrolytes contaminated with emulsified oil phase (Scheme 1E). The prepared superaerophobic coatings on NF displayed the best HER and OER performance in alkaline condition, depending on the nature of embedded ionic moieties (Scheme 1F). The influence of microenvironments for the electrochemical reactions dominates over the force of gas bubble adhesion for a capillary force-dominated gas bubble departure from working electrode. Integration of such coated NFs into a two-electrode system as working and counter electrodes reduced the required potential for overall water splitting by 500 ± 6 mV as compared to bare NF to sustain a current density of 100 mA cm^{-2}.

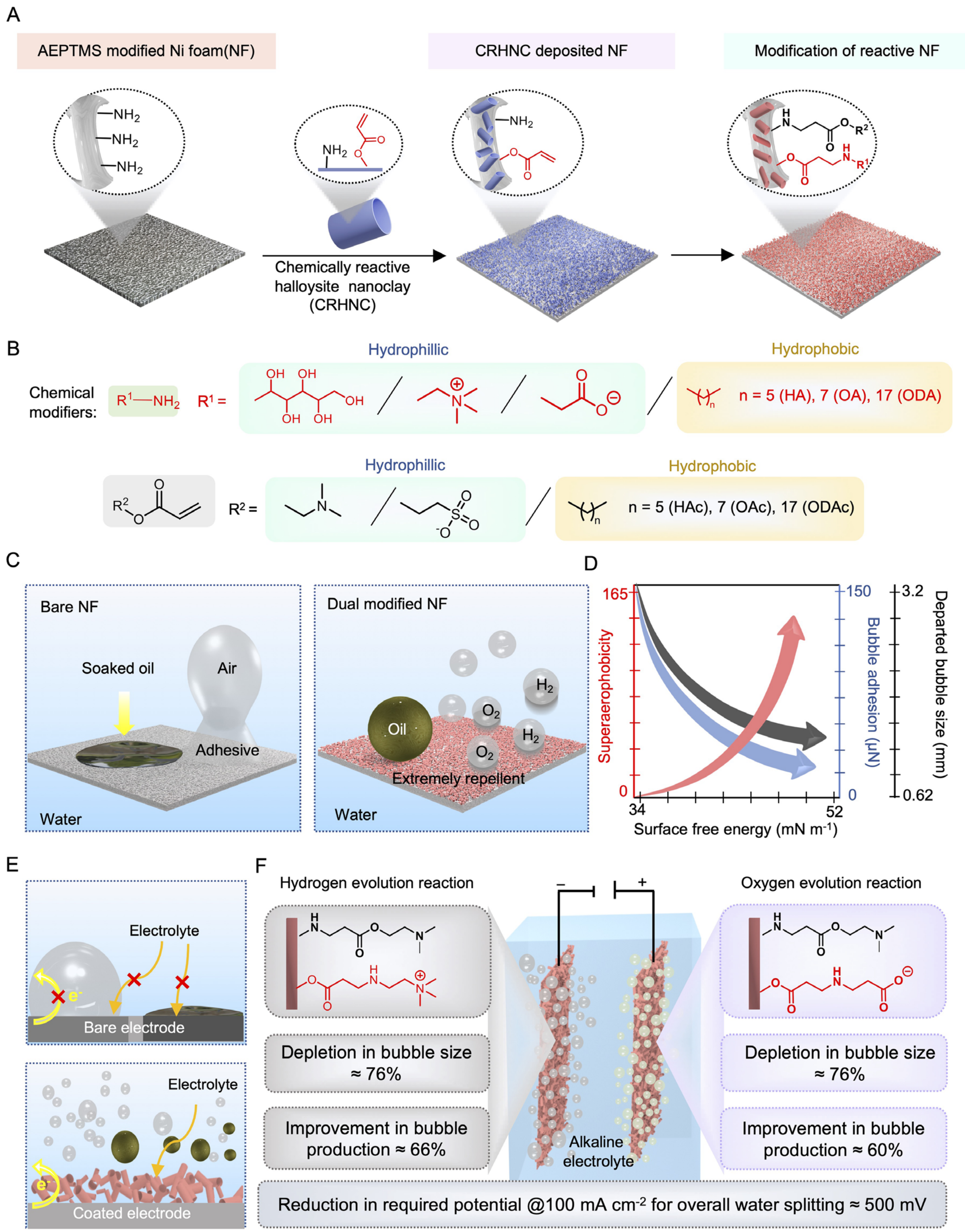
2. Results and Discussion

2.1. Halloysite Nanoclay-Based Reactive and Porous Coating and Its Post-Covalent Modifications

We selected a naturally abundant and inherently hollow nanomaterial, i.e., HNC for improving overall water-splitting through strategic association of extreme bubble repellence decorated with favorable microenvironments at the electrode surface. First, the HNC was surface-functionalized with a reaction mixture of 3-(2-aminoethylamino)-propyltrimethoxysilane (AEPTMS) and dipentaerythritol penta-acrylate (5Acl) to derive chemically reactive HNC (denoted as CRHNC) having residual primary amine and acrylate groups (Figure S1, Supporting Information). It retained the intrinsic hollow tubular structure as confirmed with field emission transmission electron microscopy (FETEM) (**Figure 1A**). The presence of amine and acrylate moieties is characterized with attenuated total reflectance Fourier transform infrared (ATR-FTIR) and X-ray photoelectron spectroscopy (XPS) analysis (Figure 1B,C). The presence of a prominent absorption peak at 1730 cm^{-1}, corresponding to C═O stretching, along with an IR signature at 1410 cm^{-1} associated with vinylic C–H stretching, confirms the presence of reactive acrylate groups in the CRHNC. Further, deconvoluted XPS spectra of reactive HNC confirms a prominent signature for primary amine at 398.9 eV. Even, the post modification of reactive HNC with the selected primary amine containing small molecules (i.e., glucamine (Glu), β-alanine (β-Ala), (2-aminoethyl)trimethylammonium chloride hydrochloride (AETMAC)) selectively depleted IR peak intensity at 1410 cm^{-1}. This effect arises from the mutual 1,4-conjugation reaction between acrylates of CRHNC and primary amine of selected small molecules, confirming selective reactivity of associated acrylate moieties on HNC toward amine groups. On the other side, we noticed a distinct change in the signatures for primary and secondary amine once CRHNC was modified with different acrylate group containing molecules as shown in Figure 1C, indicating the reactivity of the primary amine groups present on the CRHNC surface toward the acrylate functionality.

Thereafter, we deposited CRHNC onto NF electrodes premodified with AEPTMS to achieve a reactive and porous coating (Figure 1D) where more deposition of CRHNC led to blockage of the intrinsic porous structure of the NF (Figure S2, Supporting Information). We controlled the deposition of coating by adjusting the concentration of HNC in the reaction mixture (ranging from 0.005 to 0.1 M). A concentration of 0.01 M of HNC resulted in the formation of a uniform coating without obstructing the intrinsic pores of the NF. Elemental mapping analysis confirmed the homogeneous distribution of the CRHNC layer on the NF (Figure S3, Supporting Information). Following its deposition, coated NF, having residual reactivity as verified from FTIR analysis (Figure S4, Supporting Information), allowed strategic modifications with various selected modifiers to associate different types of hydrophilic and hydrophobic moieties. The modification of coated NF with primary amine-containing hydrophilic molecules—including Glu, β-Ala, AETMAC resulted in single modified (SM) coatings, referred to as SM$_{I}$, SM$_{II}$, and SM$_{III}$ respectively. Further, we modified the SM coatings individually and subsequently with different acrylate-containing modifiers like 3-sulfopropyl acrylate potassium salt (Sul.ac.; anionic), 2-(dimethylamino)ethyl acrylate (DMAEA, neutral) to obtain different dual modified (DM) coatings designated as DM$_{I}$ (β-ala/DMAEA), DM$_{II}$ (AETMAC/DMAEA) and DM$_{III}$ (AETMAC/Sul.ac.) (Figure 1E). Similarly, to obtain dual hydrophobic modifications, we modified the prepared coating with selected alkyl amines and alkyl acrylates, and such modified coatings are denoted as DM$_{IV}$, DM$_{V}$ and DM$_{VI}$ (See Figure 1E). Surface modifications were confirmed by attenuated total reflectance Fourier transform infrared (ATR-FTIR) spectroscopy (Figure S4, Supporting Information).

Thereafter, we evaluated the water and gas (hydrogen and oxygen) bubble wettability of the reactive and modified (hydrophilic and hydrophobic) coatings deposited on NF by measuring their



Scheme 1. A) Representative schematic for the deposition of chemically reactive halloysite nanoclay (CRHNC) comprised of reactive amine and acrylate groups on amine-embedded nickel foam (NF) and its post-covalent dual modification. B) Chemical structures of different modifiers: acrylate and amine groups based hydrophilic and hydrophobic molecules. C) Schematic illustrating the affinity of the bare NF toward oil droplets and gas bubbles, however, after the dual post-modification with hydrophilic moieties of the CRHNC deposited NF exhibits extreme bubble and oil repellency underwater. D) Accounting the change in different co-relevant properties (superaerophobicity, bubble adhesion and bubble size departure) with surface free energy. E) Schematic representation of gas evolution process in oil-contaminated electrolyte for bare and coated electrode where gas bubble and oil block the active sites of bare NF perturbing the performance and coated electrode has negligible perturbation due to extreme oil and bubble repellency. The drak-grey and black regions on the bare electrodes represent blocked sites by adhered gas bubble and crude oil, respectively. F) Showing performance of different dual modifications of the prepared coating toward hydrogen evolution reaction (HER) and oxygen evolution reaction (OER).

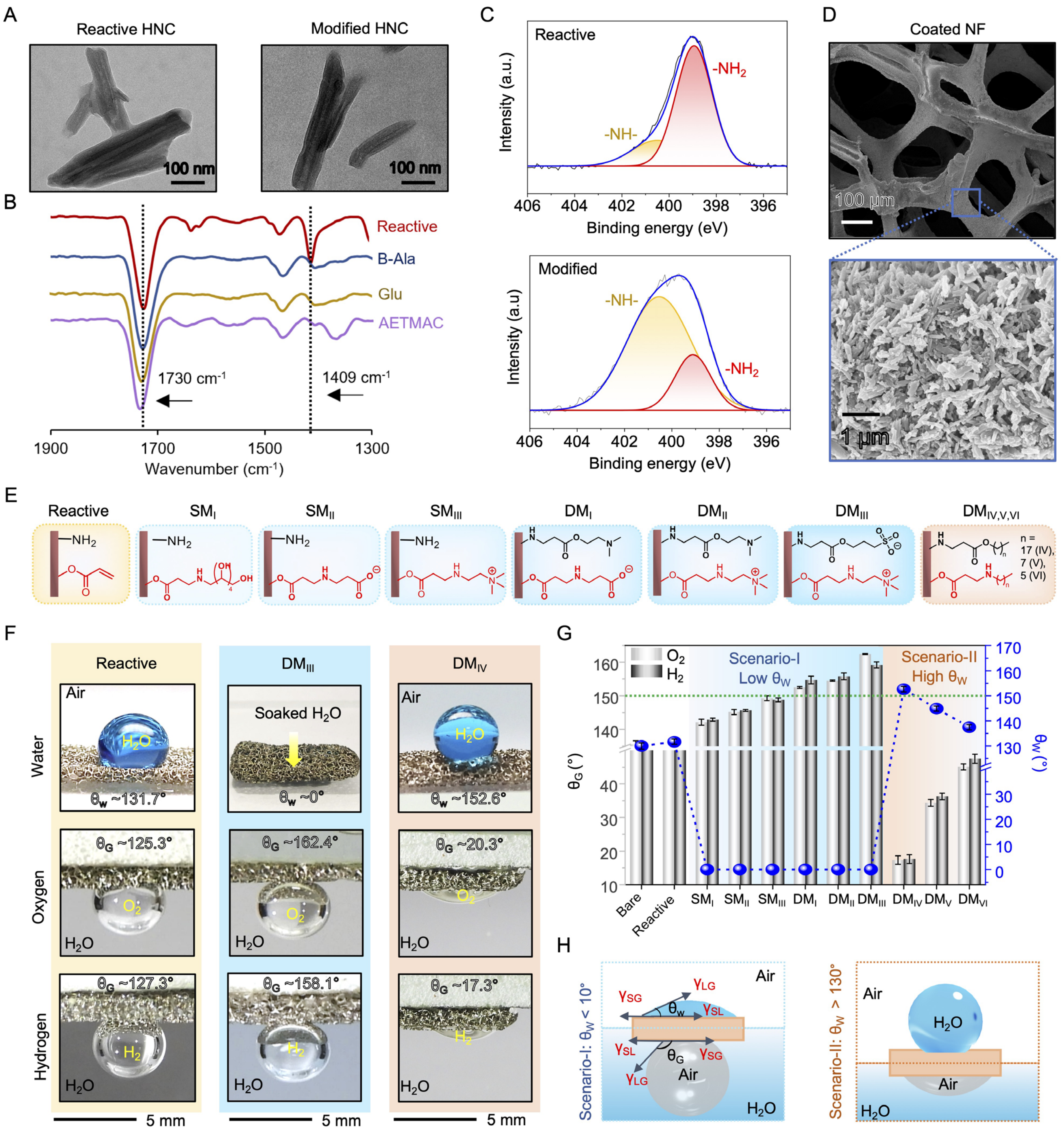


Figure 1. A) Field emission transmission electron microscopic (FETEM) images of reactive halloysite nanoclay (HNC) powder having residual reactive groups before (left panel) and after its post-modification (right panel). B) Attenuated total reflection Fourier transform infrared (ATR-FTIR) spectra of chemically reactive HNC and different single-modified HNC. C) X-ray photoelectron spectroscopy (XPS) spectra of CRHNC and dual-modified HNC powder. D) Field emission scanning electron microscopic (FESEM) images of the NF after depositing the CRHNC at low and high magnifications. E) Illustrating the chemical functionality on the coated NF before and after different single and dual modifications. F) Digital images showing water and bubble wettability on reactive (left panel), DM_{III} (middle panel) and DM_{IV} (right panel) coated NFs. G) Accounting the influence of different single (hydrophilic) and dual (hydrophobic and hydrophilic) modifications on underwater gas bubble contact angle [θ_G (hydrogen and oxygen)] and water contact angle (θ_w). Different hydrophilic modifications leads to scenario-1(low θ_w) and hydrophobic modifications leads to scenario-2 (high θ_w). H) Schematic illustration of the equilibrium θw and θ_G for hydrophilic surface (Scenario-I) and hydrophobic surface (Scenario-II), where the interfacial forces acting on the water droplet in air and gas bubble underwater at the three-phase contact point. The error bar indicates the standard deviation with number of measurements, $n = 3$ for each data point.

contact angles (denoted as θ_W for water in air and θ_G for gas bubbles underwater) (Figure 1F; Figures S5 and S6, Supporting Information). The deposited reactive coating on NF resulted in a minimal (< 5°) change in θ_W and slightly decreased θ_G compared to bare NF (Figure 1F,G). However, the association of selected hydrophilic modifications (SM_I to DM_{III}) rendered the surfaces with superhydrophilicity (θ_W of 0°; Figure 1G; Figure S5, Supporting Information), where θ_G remains either >150° (indicating superaerophobicity) or <150° depending on the selection of single or dual modifications. As per the Young's wettability model, $\theta_G + \theta_W = 180°$ on a smooth solid interface.[27,28] The same interfacial forces are acting on beaded water droplets in air and gas bubbles underwater (Figure 1H), where superhydrophilic interfaces are supposed to display superaerophobicity. In our current study, some dual-modified porous coatings ($DM_{I\,-}$ DM_{III}) provided superaerophobicity with θ_G >150°, however, the modification of the same coating with alkyl moieties ($DM_{IV\,-}$ DM_{VI}) of higher analogues of hexyl moieties resulted in a gradual decrease in θ_G (Figure 1G; Figure S6, Supporting Information). Thus, the current method of post-modifying prepared reactive and porous coating on NF with selected hydrophobic and hydrophilic modifiers provided a broad spectrum of gas-bubble wettability to examine its role in electrochemical overall water-splitting. During the process of post-modifications with different chemical modifiers, the coating morphology remained very similar, suggesting chemical modifications mainly remain responsible for tailoring gas bubble wettability (Figure S7, Supporting Information). Notably, in this current study, both single and dual-modified coatings on NF exhibit superhydrophilicity ($\theta_W \sim 0°$)—but some of them failed to display superaerophobicity (Figure 1G). This observation leads to a critical question: can hydrophilicity alone be considered as a reliable predictor of superaerophobicity at solid–liquid–gas interfaces?

To understand this unusual gas-bubble wetting behavior, we analyzed the relationship between the apparent gas-bubble contact angle (θ_a) measured on differently-modified porous coatings on NF and Young's gas-bubble contact angle (θ_Y) determined on equivalent smooth coatings on glass substrate. A linear increase of θ_a with θ_Y was observed, which can be fitted into following equation, i.e., $\cos\theta_a = 1.7\cos\theta_Y + 0.4$, with a high fitting reliability ($R^2 = 0.99$) as shown in **Figure** 2A. But, it follows a different trend while compared with other classical wetting models—including the Wenzel model ($\cos\theta_a = r\cos\theta_Y$) and Cassie–Baxter model ($\cos\theta_a = \varphi\cos\theta_Y - (1-\varphi)$), where r represents surface roughness of solid and φ is the solid–bubble contact fraction ($0 < \varphi < 1$).

Classical wettability models typically predict a negative or zero intercept on the $\cos\theta_a$-axis,[29–31] the current experimental results provide a positive intercept of 0.4 as shown in Figure 2A. Hence, it is essential to reconsider the current understanding on gas-bubble wetting phenomena. Since in the current study, morphology of the coating on NF remained nearly unaltered, we assumed that the selection of modifiers is mainly responsible for tailoring the gas bubble wettability. Hence, we attempted to examine the surface free energy of modified coatings on smooth substrates experimentally and theoretically to understand gas bubble wettability in the following section.

2.2. Impact of Surface Free Energy on Gas Bubble Wettability and Adhesion

In the last section, we noticed the modified coatings displayed superhydrophilicity with a final θ_W of 0°. But, water droplets (5 µL) initially beaded on such modified coatings with a non-zero CA and it gradually depleted to 0° as depicted in Figure 2B. We measured the transition time required by differently modified coatings to achieve a static CA of 0° and it was denoted as $t_{CA(0°)}$. For instance, $t_{CA(0°)}$ was 1000 times more (16,000 ms vs. 16 ms) for SM_I compared to DM_{III}-coated NF (Figure 2B; Figure S8, Supporting Information). To understand this behavior, we estimated the surface free energy (γ_{SG}) of all hydrophilic modified coatings experimentally and theoretically. To theoretically calculate the γ_{SG} of the surfaces, model surfaces were constructed for molecular dynamics simulation (Figure S9, Supporting Information) and γ_{SG} was calculated from the contact angles of polar and non-polar components (Figures S9–S13, Supporting Information) and the results were compared with experimentally obtained γ_{SG} (Figure S14, Supporting Information). They followed a very similar trend, and the change in γ_{SG} is mainly because of chemical modifications. We found that the $t_{CA(0°)}$ gradually decreases with an increase in γ_{SG} (Figure 2C; Figure S8, Supporting Information). Coating with high γ_{SG} promotes a better spreading of a water droplet, suggesting its improved hydration (Figure 2D). We verified this observation by molecular dynamics simulations, revealing a greater number of surface functional groups wetted in DM_{III} than in SM_I (Figure 2E). Since the number of water molecules in the droplet is same on both the surfaces, a lower coordination number dictates more spreading of the water and hence better wetting. Thus, both experimental and simulation results suggest that a better water spreading on coating having higher γ_{SG}, which attributes to appropriate chemical modifications.

Furthermore, we examined the adhesion of gas bubbles on differently modified coatings on NF both qualitatively and quantitatively. When we made a gas bubble (8 µL) in contact with the bare NF and subsequently dragged over it, a significant distortion of the bubble shape was observed because of the strong adhesion of gas bubbles (Figure 2F). In contrast, we noticed minimal or no distortion when an identical experiment was repeated on a dual-modified coating (DM_{III}), suggesting a much lower adhesion of the gas bubble (Figure 2E; Figure S15, Supporting Information). Next, we measured the adhesion force of the coatings with gas bubble using a force tensiometer, where a gas bubble (8 µL) was individually brought in contact with differently modified coatings and it receded to detach from the respective coatings. We noticed that the gas bubble adhesion force gradually depleted with increasing γ_{SG} of the coating. The coating with the highest γ_{SG} (i.e., 52.8 ± 2.1 mN m^{-1}) exhibited the lowest adhesion force of 4.7 ± 0.7 µN, due to its improved hydration, which minimizes bubble–surface interactions. Conversely, the hydrophobic modification either suffers from a significantly elevated gas bubble adhesion force or superaerophilicity allowing complete soaking of the beaded gas bubble, preventing the measurement of adhesion force (Figure 2F). Thus, chemically modulated γ_{SG} of the

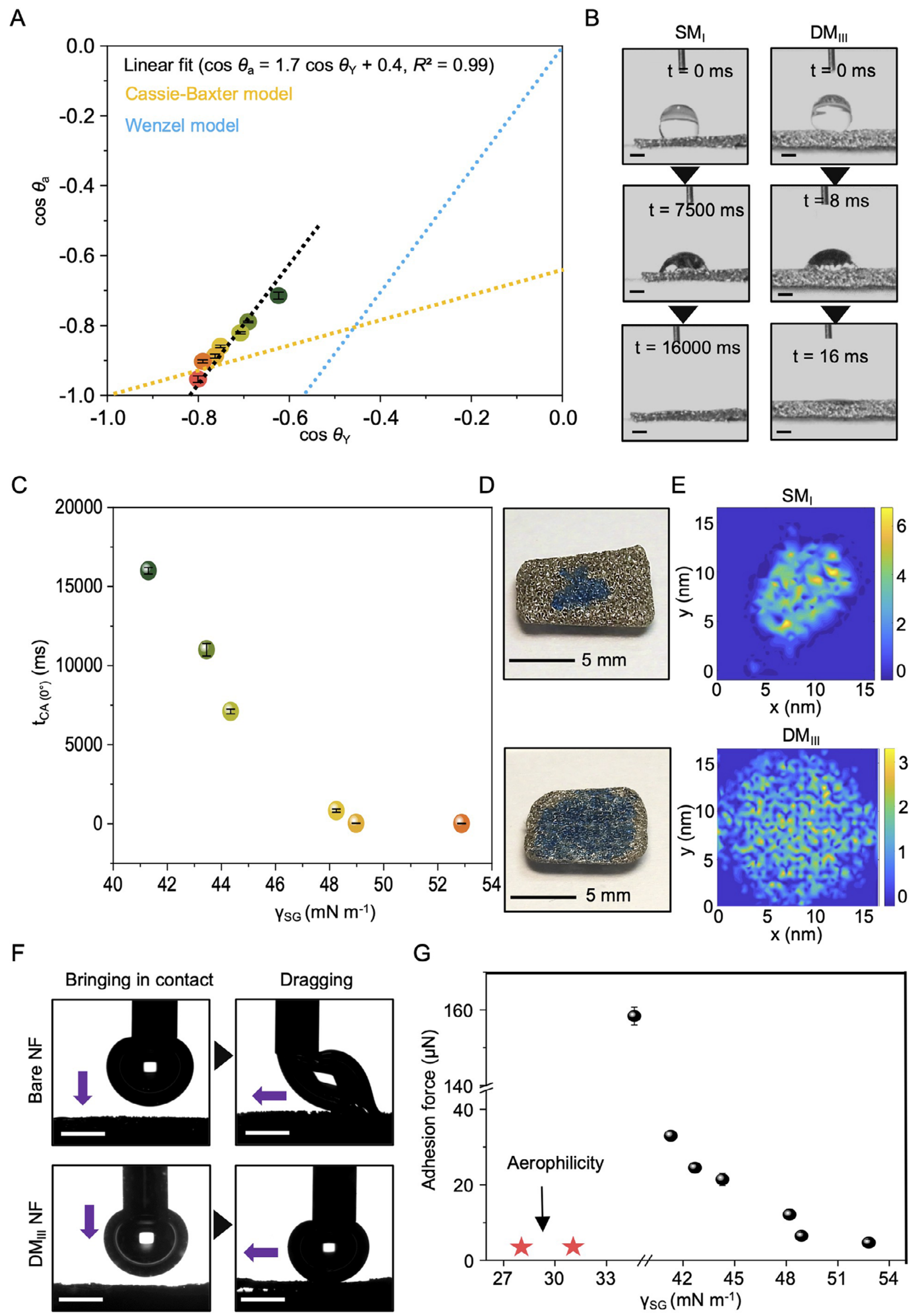


Figure 2. A) Diagram correlating the Young's gas-bubble contact angle (θ_Y) with the apparent gas-bubble contact angle on bare and coated NF (θ_a). The Cassie–Baxter (yellow dotted) and Wenzel (blue dotted) models are included for comparison. B) High-speed camera images in different time intervals illustrating the soaking process of beaded water droplet on SM_I and DM_{III}-coated NFs (Scale bar, 1 mm). C) Diagram representing the water spreading time on different single and dual (hydrophilic) modified coatings on NFs. D) Digital images showing the water soaking on SM_I and DM_{III}-coated NFs. E) 2D contour plots showing the coordination number distribution, representing the water spreading on SM_I and DM_{III}-coated NFs. The *x*- and *y*-axes correspond to the average positions of functional group atoms in the surface plane, while the *z*-axis (color scale) denotes the local coordination number. F) Contact angle goniometer images accounting for the beading and dragging of a gas bubble on bare and DM_{III}-coated NFs, where distortion of the bubble signifies high adhesion in case of bare NF and no distortion for DM_{III}-coated NF (Scale bar, 2 mm). G) Accounting for the relationship between force of bubble adhesion on bare NF, single and dual modified (hydrophilic and hydrophobic) NFs with γ_{SG}. The error bar indicates the standard deviation with number of measurements, $n = 3$ for each data point.

prepared coatings played a pivotal role in tailoring gas bubble wettability and its adhesion force. Its implication on electrochemical water splitting is discussed in the following sections. Moreover, we evaluated the chemical durability of DM_{III}-coated NF by monitoring the θ_G. For this, we subject the sample to various chemically harsh environments, and the θ_G measurements revealed that the coating retained its superaerophobicity even after 30 days of continuous and harsh chemical exposures (Figure S16, Supporting Information).

2.3. Impact of Prepared Coatings on Electrochemical Gas Evolution Reactions

In this section, we investigated the influence of CRHNC-based coatings and their post-covalent hydrophilic (DM_I to DM_{III}) and hydrophobic (DM_{IV} to DM_{VI}) modifications on the electrochemical performance, under a standard three-electrode configuration in an alkaline electrolyte (1 M KOH). While, we selected the bare and coated (with and without post-covalent modifications) NF, Hg/HgO electrode and a graphite rod as the working electrode, reference electrode and counter electrode respectively. For HER, we conducted measurements over a potential range of 0 to -0.9 V versus the reversible hydrogen electrode (RHE) (**Figure 3A**). In contrast, we studied OER within a potential range of 1.2 to 1.9 V versus RHE (Figure 3B). We examined the HER and OER activities by studying linear sweep voltammetry (LSV). Following the deposition of reactive coating on NF, both HER (Figure 3C) and OER (Figure 3D) performance declined. In contrast, hydrophilic dual-modifications with different surface charge functionalities (DM_I: anionic DM_{II}: cationic DM_{III}: zwitterionic) led to improved HER and OER performance as compared to bare NF (Figure 3C,D; Figure S17, Supporting Information). But a completely different trend is observed when hydrophobic modifications (DM_{IV} to DM_{VI}), are associated (Figure 3C,D; Figure S18, Supporting Information). To understand this behavior, we attempted to correlate the γ_{SG} of the unmodified and modified coatings with overpotential for HER and OER, at a current density of 10 mA cm^{-2}. We observed an initial depletion in overpotential for HER and OER with increasing γ_{SG} (Figure 3E,F). Because a higher γ_{SG} improves hydration of the surface, reducing the adhesion force between gas bubbles and the electrode surface, thereby promoting the detachment of smaller bubbles to maintain a nearly unperturbed access of electrolytes to catalytically active sites (Movie S1, Supporting Information). However, modified coating with the highest surface free energy (i.e., DM_{III}) did not display the best electrochemical gas evolution performance. To further understand this observation, we calculated the Eötvös number (Eo), a dimensionless parameter accounting for the relative effect of buoyancy force against capillary forces, using the given equation:

$$\mathrm{Eo} = \left(\rho_W - \rho_G\right) g D_b^2 \gamma_{\mathrm{WG}}^{-1} \tag{1}$$

where, ρ_W and ρ_G are the density of water and gas respectively, g is the gravitational acceleration constant, γ_{WG} represents the water-gas interfacial tension and D_b is the diameter of the detaching gas bubble. Typically, high Eo, suggests buoyancy force-dominated bubble departure from reactive and modified coatings (DM_{IV} to DM_{VI}) because of strong gas bubble adhesion and low γ_{SG}. On the other side, hydrophilic modifications of the same coating conferred significantly low Eo, ensuring a capillary force-dominated bubble departure as a high γ_{SG} results in a low force of bubble adhesion on the modified coatings. It is worth mentioning that both bare and coated NF with hydrophobic modifications displayed a significantly depleted Eo at elevated temperatures (<100 °C), indicating a transition from buoyancy force dominance to capillary force-governed gas bubble detachment (Figure 3I). Modified NFs with high γ_{SG} inherently exhibit low bubble adhesion forces, facilitating capillary-dominated bubble detachment even at lower temperatures. Thus, γ_{SG} plays a crucial role in governing bubble repellence and adhesion force at ambient conditions. The electrochemical gas evolution reactions by currently prepared coatings with high γ_{SG} (>48 mN m^{-1}) displayed an unusual trend, indicating the existence of some other factors beyond bubble repellence and bubble adhesion for depleted overpotential for HER and OER. We noticed a modified coating with cationic moiety (DM_{II}) exhibited the highest HER activity, with a current density of −1185 mA cm^{-2} at −0.9 V versus RHE and the lowest overpotential of 138 mV versus RHE to reach a current density of 10 mA cm^{-2} (Figure 3C,E). Another coated NF decorated with the anionic moiety (DM_I) conferred the best OER performance, with a current density of 550 mA cm^{-2} at 1.9 V versus RHE and a minimum overpotential of 290 mV versus RHE to reach 10 mA cm^{-2} (Figure 3D,F). Thus, modified coatings with high γ_{SG}, associated with capillary force-dominated gas bubble departure, displayed best HER and OER performance depending on the embedded ionic moieties, where high γ_{SG} and appropriate ionic environments of the modified coating synergistically improved electrochemical gas evolution reaction, likely through improved reaction kinetics and maintaining unperturbed access of catalytically active sites.

In the past, association of positively charged surface groups in the vicinity of catalytically active sites facilitated the HER by stabilizing transition states in the Volmer step, thereby enhancing the reaction kinetics.[15] Inspired by this seminal result, we hypothesized that because of the decoration of the superaerophobic coating with quaternary amine, the positive zeta potential (**Figure 4A**) facilitates the kinetics of water dissociation as schematically depicted in Figure 4B. To understand this observation, we employed density functional theory (DFT) to compute the Gibbs free energy change (ΔG^o_{WD}) of water dissociation by the surface molecules of the dual modified coatings (Figure S19, Supporting Information). A more negative ΔG^o_{WD} (Table S1, Supporting Information) indicates stronger ability for polarizing water molecules (further confirmed by molecular electrostatic potential (MESP) map in Figure S20, Supporting Information) at electrode surface, particularly for coatings functionalized with quaternary amine groups, and thus a better catalytic potential is expected with faster HER kinetics (Table S1, Supporting Information and Figure 4B). To decouple the individual contributions of bubble repellency and ionic microenvironment on HER performance, we designed two control surfaces systematically. To isolate the effect of the ionic environment, we used a non-ionic glucamine group in place of AETMAC, while retaining DMAEA as the second modification (Control-I, Figure 4D). This surface maintained superaerophobicity (Figure 4E) but lacked a cationic microenvironment. Conversely, to examine the role of bubble repellency, we employed

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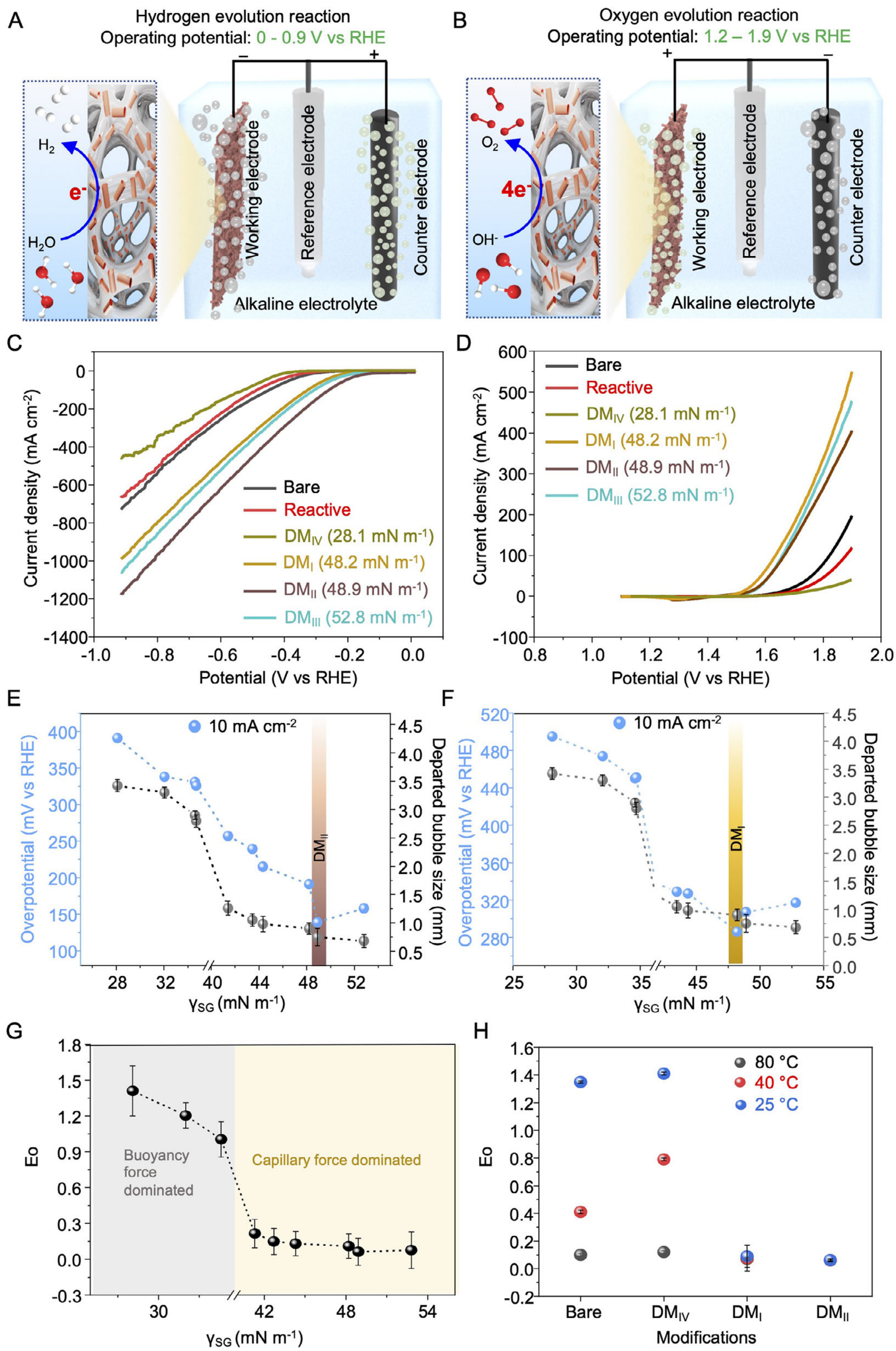
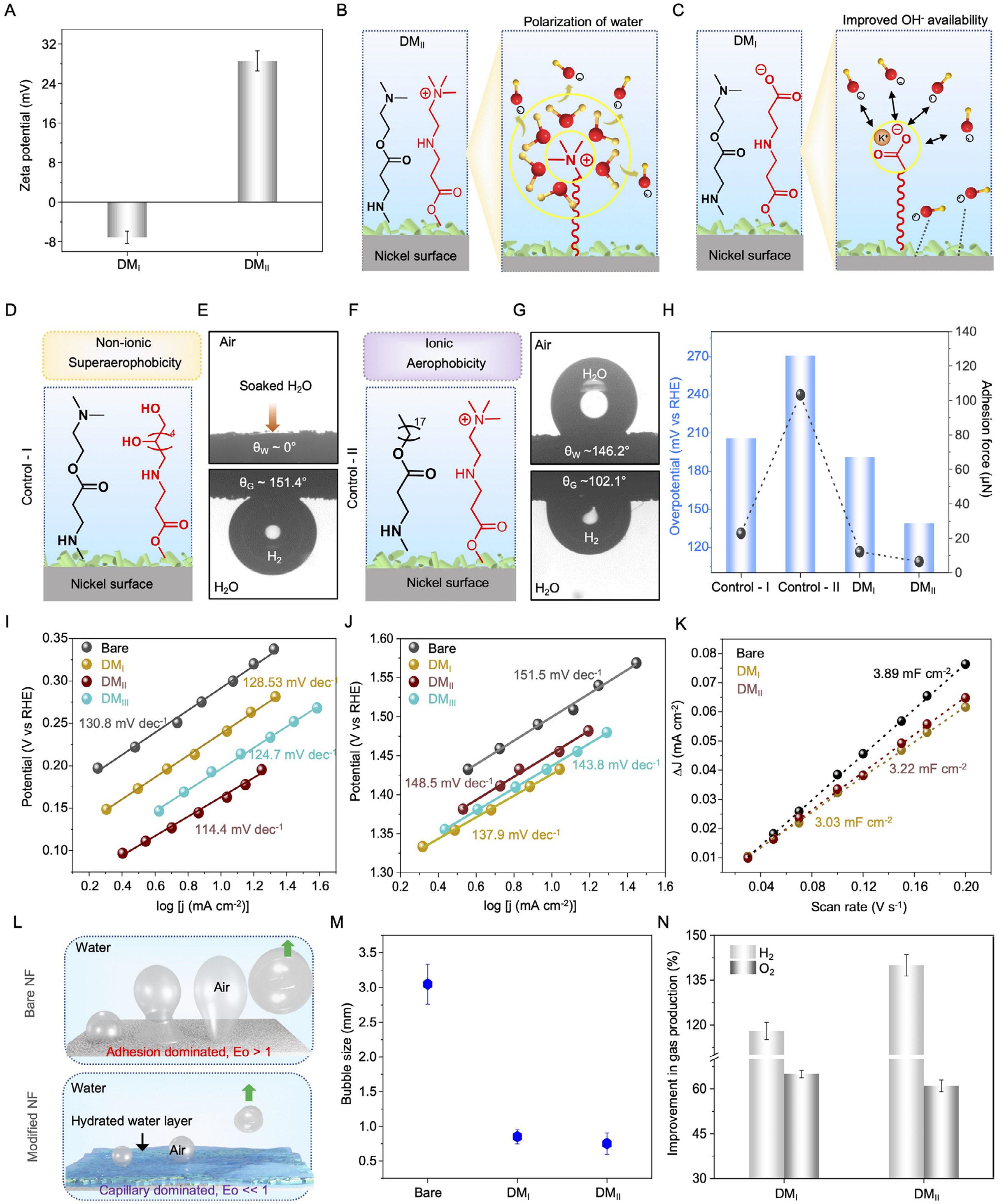


Figure 3. A,B) Schematic illustration of the hydrogen (A) and oxygen (B) evolution reactions. C,D) Linear sweep voltametric (LSV) plots of bare, reactive and differently modified NFs for HER (C) and OER (D) in 1 M KOH. E,F) Diagram representing the relationship between overpotential (at 10 mA cm^{-2}) and departed bubble size with γ_{SG} during HER (E) and OER. (F). G) Change in Eotvos number (E_o) with $\gamma_{SG.}$ For surfcae with low $\gamma_{SG,}$ bubble departure is dominated by buoyancy and surfaces with high γ_{SG} display capillary force dominated bubble departure. H) Effect of temperature on Eo for bare and differently modified NFs. The error bar indicates the standard deviation with number of measurements, $n = 3$ for each data point.


A
Zeta potential (mV)
32
24
16
8
0
-8
DM_I
DM_{II}
B
DM_{II}
Polarization of water
Nickel surface
C
DM_I
Improved OH^- availability
Nickel surface
D
Non-ionic Superaerophobicity
Control - I
Nickel surface
E
Air
Soaked H_2O
$\theta_W \sim 0°$
$\theta_G \sim 151.4°$
H_2
H_2O
F
Ionic Aerophobicity
Control - II
Nickel surface
G
Air
H_2O
$\theta_W \sim 146.2°$
$\theta_G \sim 102.1°$
H_2
H_2O
H
Overpotential (mV vs RHE)
Adhesion force (µN)
Control - I
Control - II
DM_I
DM_{II}
I
Bare
DM_I
DM_{II}
DM_{III}
130.8 mV dec^{-1}
128.53 mV dec^{-1}
124.7 mV dec^{-1}
114.4 mV dec^{-1}
Potential (V vs RHE)
log [j (mA cm^{-2})]
J
Bare
DM_I
DM_{II}
DM_{III}
151.5 mV dec^{-1}
148.5 mV dec^{-1}
143.8 mV dec^{-1}
137.9 mV dec^{-1}
Potential (V vs RHE)
log [j (mA cm^{-2})]
K
Bare
DM_I
DM_{II}
3.89 mF cm^{-2}
3.22 mF cm^{-2}
3.03 mF cm^{-2}
ΔJ (mA cm^{-2})
Scan rate (V s^{-1})
L
Bare NF
Water
Air
Adhesion dominated, Eo > 1
Modified NF
Water
Hydrated water layer
Air
Capillary dominated, Eo << 1
M
Bubble size (mm)
Bare
DM_I
DM_{II}
N
H_2
O_2
Improvement in gas production (%)
DM_I
DM_{II}

AETMAC to introduce a cationic environment, while the second modification was replaced with ODAc, resulting in Control-II (Figure 4F). This surface possessed a cationic character but failed to exhibit superaerophobicity (Figure 4G). Electrochemical measurements revealed that both Control-I and Control-II resulted a higher overpotentials to sustain the same current density compared to the DM_{II}-coated NF (Figure 4H). Similarly, to investigate the individual role of an anionic microenvironment, we designed control surfaces using β-alanine-based systems (Figure S21, Supporting Information). These findings indicate that neither bubble repellency nor ionic microenvironment alone is sufficient to substantially improve HER/ OER performance. Instead, their synergistic integration is essential for optimal electrocatalytic performance.

Then, we compared Tafel slopes for best performing coatings (DM_I and DM_{II}) having high γ_{SG} and distinct ionic environments. Coated NF decorated with cationic moieties exhibited a lower Tafel slope of 114.4 mV dec^{-1} for HER compared to 130.8 mV dec^{-1} for bare nickel foam, indicating more favorable HER kinetics (Figure 4I). Similarly, for OER, we noticed improved kinetics with anionic moieties. We hypothesize that surfaces with negatively charged groups repel OH^- ions from the bulk electrolyte, improving the availability of OH^- ion at the electrode surface and promoting more efficient oxygen evolution (Figure 4C). To investigate the role of ionic environment provided by prepared superaerophobic coating on availability of OH^- ions during OER, we have estimated the ion diffusion coefficient (D) using the Warburg coefficient (σ) following the relation:[32,33]

$$D = 0.5 \times \frac{(RT)^2}{\left(An^2 F^2 C\sigma\right)^2} \tag{2}$$

here, R denotes the gas constant (8.314 J mol^{-1} K^{-1}), T is the absolute temperature (298 K), A represents the electrode surface area, n is the number of electrons involved in the electrochemical reaction, F is the Faraday constant (96,500 C mol^{-1}), C is the concentration of OH^- ions in the electrolyte (1 M KOH), and σ corresponds to the Warburg coefficient, which is determined from the slope of the linear fit between the real part of impedance (Z′) and the inverse square root of angular frequency ($\omega^{-0.5}$) (Figure S22, Supporting information). We noticed ≈20 times higher D (5.58×10^{-19} vs. 2.66×10^{-20} cm^2 s^{-1}) for superaerophobic coating decorated with anionic moiety in comparison to bare NF. A higher D value indicates faster OH^- ion transport at electrode surface for an efficient charge transfer during OER.

Thereafter, we noticed a lower Tafel slope of 137.9 mV dec^{-1} for superaerophobic coating having negative zeta potential (Figure 4A) in comparison to bare and other coated NF, highlighting a positive impact of anionic surface chemistry on oxygen evolution (Figure 4J). We confirmed this hypothesis further by DFT calculations (Figure S14, Supporting Information). The least negative value of $\Delta G^0_{\mathrm{Comp}}$ for DM_I (β-Ala – DMAEA) surface indicates that this surface is less prone to interact with free OH^- ions and, therefore, more likely to promote the OER (Table S1, Supporting Information).

Thereafter, we evaluated the electrochemically active surface area (ECSA) for both bare and coated NFs from cathodic and anodic current density difference (ΔJ) versus scan rate plot in Figure 4K. The double-layer capacitance (C_{dl}) values were 3.89, 3.22, and 3.03 mF cm^{-2} for bare NF, DM_{II} and DM_I-coated NF, respectively. Thus, the deposition of electrically insulating coatings (DM_I and DM_{II}) on NF depletes ECSA with respect to bare NF (Figure 4K; Figure S23, Supporting Information). However, these results should not be interpreted in isolation as it was measured in the non-Faradaic potential range,[34,35] where no charge transfer occurs for gas evolution. In the Faradaic region, where gas evolution actually takes place, the behavior of these surfaces differs substantially. On bare nickel foam, strong adhesion of gas bubbles causes them to linger on the surface, requiring them to grow larger before buoyancy-driven detachment can occur. Eventually, a delayed detachment of produced gas bubbles on bare NF temporarily blocks catalytic sites (Figure 4L). In contrast, the coated NF having high γ_{SG} ensures a hydrated interface, reducing bubble adhesion significantly. As a result, even small gas bubbles detach easily from the surface, keeping catalytically active sites more accessible for performing HER as schematically depicted in Figure 4L. In fact, the average detached bubble size on bare nickel foam was observed to be 2.95 ± 0.28 mm, while much smaller bubbles continued to detach from DM_I (0.9 ± 0.10 mm) and DM_{II} (0.75 ± 0.15 mm) coated NF (Figure 4M). This improved bubble dynamics leads to more effective use of the catalytic surface during their GERs. Although the apparent ECSA in the non-faradaic region is lower for DM_I and DM_{II}, the functional active area during gas evolution is effectively higher. We confirmed this further by gas collection experiments under identical electrochemical conditions, where potential of −0.55 and 1.7 V versus RHE were applied to perform HER and OER. We noticed NFs coated with DM_I and DM_{II} are capable of producing >60% more gas than bare NF (Figure 4N) during HER and OER respectively. Then, we examined the electrochemical stability of

Figure 4. A) Diagram depicting the zeta potential on the different dual-modified HNC surfaces. B,C) Schematic illustration of the different ionic environments by modified coating on HER and OER, cationic modification helps in the polarization of water (B) and anionic modification ensures the freely available OH^- near the electrode proximity (C), verified by experiment and supported by free energy calculations using DFT. D) Depicting the non-ionic superaerophobicity termed as Control–I. E) Contact angle goniometer images illustrating water contact angle (top panel) and gas bubble contact angle (bottom panel) for Control–I. F) Depicting the ionic aerophobicity termed as Control–II. G) Contact angle goniometer images illustrating water contact angle (top panel) and gas bubble contact angle (bottom panel) for Control–II. H) Diagram comparing the overpotential for HER at 100 mA cm^{-2} and gas bubble adhesion force for Control–I, Control–II, DM_I and DM_{II}-coated NFs. I,J) Tafel plots of bare and coated (DM_I, DM_{II} and DM_{III}) NF for HER (I) and OER (J). K) Cathodic and anodic current density difference (ΔJ) versus scan rate plot of bare, DM_I and DM_{II}-coated NF. L) Schematic representing the interaction of the substrate and bubble during gas evolution process for bare NF and DM NF. For, adhesive surface, large bubbles departing during GER, whereas coating with high γ_{SG} ensures the release of smaller gas bubbles. M) Comparison of the departed bubble size during gas evolution in case of bare, DM_I and DM_{II}-coated NF. N) Diagram showing the improvement in gas (H_2 and O_2) production by coated NFs with respect to bare NF in a three electrode experimental set-up in alkaline condition (1 M KOH) at −0.55 V versus RHE and 1.7 V versus RHE respectively, where DM_I and DM_{II}-coated NFs were individually selected as working electrode. The error bar indicates the standard deviation with number of measurements, $n = 3$ for each data point.

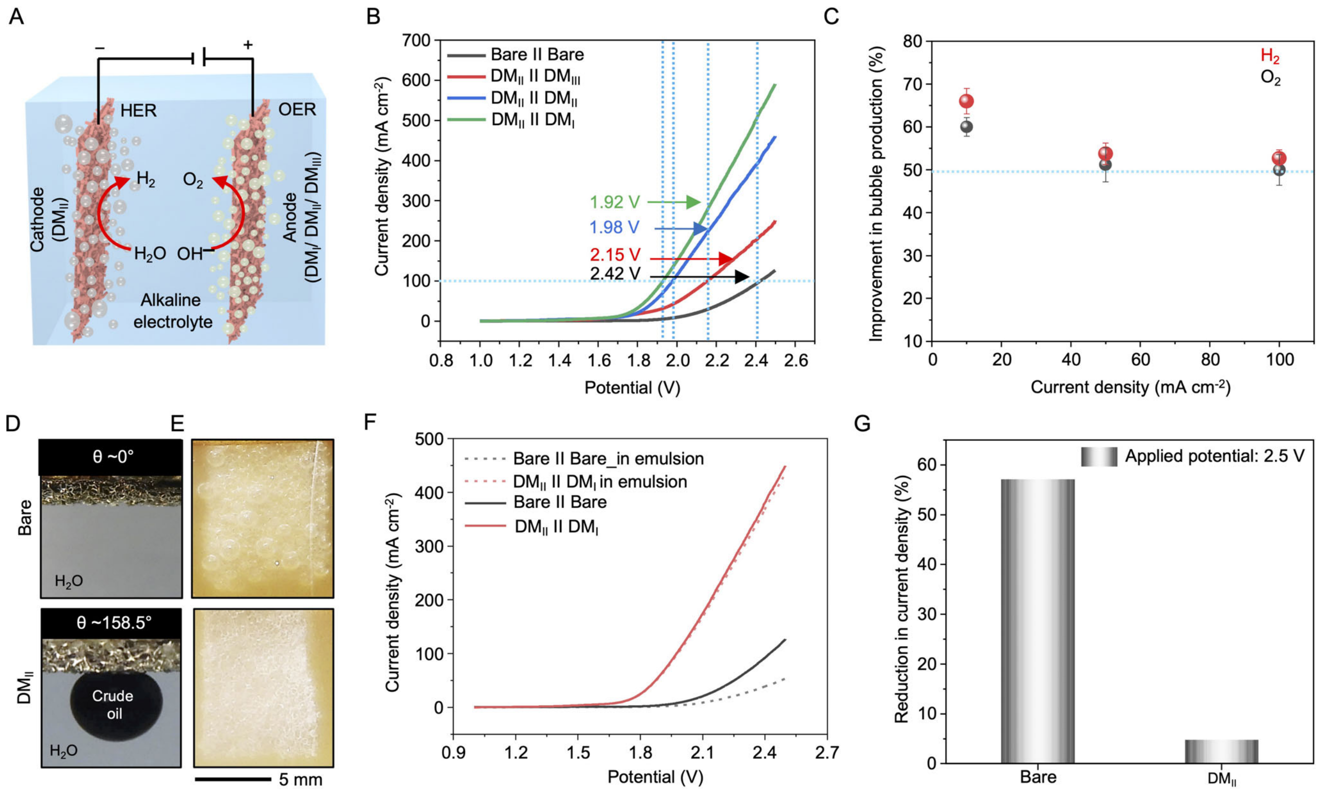


Figure 5. A) Schematic representation of the two-electrode set-up for overall water-splitting where DM_{II}-coated NF is employed as the cathode and DM_I, DM_{II}, DM_{III}-coated NF are employed as anodes. B) LSV traces for overall water splitting by bare and differently modified NFs. C) Displaying the improvement in bubble production in two-electrode set up by coated NFs (DM_{II} and DM_I coated NFs are selected as cathode and anode) with respect to bare NF for hydrogen (red) and ogygen (balck) gases. D) Digital images illustrating the superoleophilicity by bare NF (top panel) and superoleophobicity by DM_{II}-coated NF (bottom panel). E) Gas evolution process of bare and DM_{II}-coated NF in oil-contaminated electrolyte. F) Comparison of LSV traces for bare and modified NFs recorded in 1 M KOH and oil-contaminated electrolytes. While the bare NF exhibits significant performance degradation in the presence of oil contamination, the coated NFs have a negligible effect. G) Reduction in current density at an applied cell-voltage of 2.5 V for bare and DM_{II}-coated NF in the presence and absence of oil-contaminated electrolyte. The error bar indicates the standard deviation with the number of measurements, $n = 3$ for each data point.

these coatings (DM_I and DM_{II}) at 250 and −250 mA cm^{-2} for HER and OER respectively. Both coatings maintained a stable performance for at least 20 h. Further, we confirmed the durability of the coating by comparing the LSV traces before and after the stability test where a negligible change in the LSV was observed (Figures S24 and S25, Supporting Information). The current method of tailoring ionic microenvironment and bubble repellence provided a facile basis for improving both HER and OER with respect to various electrochemical parameters, including overpotential, current density and Tafel slope, without associating any external catalyst in the coating. Moreover, it exhibits a superior bifunctional performance while compared with recently reported relevant literatures (Table S2, Supporting Information).

2.4. Water Splitting in Oil-Contaminated Electrolyte by Coated NF

To explore the practical applicability of prepared coatings, we employed a two-electrode cell configuration to demonstrate overall water splitting (**Figure** 5A). In this setup, we used DM_{II}-coated NF as the cathode and DM_I, DM_{II} and DM_{III}-coated NF were applied separately as anodes. Among all combinations, the pairing of DM_{II}-coated NF as the cathode and DM_I-coated NF as the anode delivered the best overall water-splitting performance (Figure 5B; Figure S26, Supporting Information), underscoring the synergistic effect of surface charge modulation and extreme gas bubble repellence on both HER and OER. We compared the bubble production rate using this two-electrode set-up for bare and coated (DM_I and DM_{II}) NF at different current densities.[16] Selected coating combination improved the gas production by ≈50% at a current density of 100 mA cm^{-2} (Figure 5C). Notably, the modified surfaces enable rapid detachment of even small bubbles, preventing blockage of catalytic sites and maintaining high activity during their performance for GER. These coated NFs also exhibit remarkable oil repellence, due to the formation of a highly hydrated interface. For instance, DM_I and DM_{II} displayed superoleophobicity with CA for a beaded crude oil droplet of 156.9 ± 2.1° and 158.5 ± 1.5° respectively, whereas bare NF immediately absorbs the oil droplet upon contact (CA ~ 0°), indicating high oleophilicity (Figure 5D). When bare NF is immersed in an oil-in-water emulsion, dispersed oil droplets readily adhered and followed by rapidly spread on it. Even after its re-immersion in fresh water, it fails to remove strongly adhered oil, leaving the surface persistently contaminated as visually evident from its brown coloration (Movie S2, Supporting Information). In contrast, the DM_{II}-coated NF has the ability to keep it free from oil contamination (Movie S2, Supporting Information). Such self-cleaning ability has significant implications for water electrolysis of waste-water from oil refineries or other industries involved in heavy usage of oil or oily liquids. Thereafter, we performed overall water splitting in oil-contaminated electrolytes (Figure 5E). The electrochemical gas evolution reaction performance by bare NF deteriorated noticeably, while the performance of coated NF remained mostly unaltered at an identical experimental set-up. Specifically, at an applied potential of 2.5 V, bare NF exhibited a reduction in current density of 57%, indicating substantial compensation of its performance because of the blockage of electrode surface by adhered oil phase. In comparison, the modified electrodes displayed <5% of change in current density at identical experimental condition, because of its exceptional resistance to fouling by oil phase and sustained catalytic performance under adverse conditions (Figure 5F,G). Thus, we have successfully depicted the strategic choice of post-covalent modification of a non-catalytic coating for improving overall water splitting performance. Further, association of other low-cost catalysts with such porous clay-based coating is likely to provide a facile basis for maximizing the catalytic performance for electrochemical water splitting.

3. Conclusion

In this study, we report a design of CRHNC to derive a bifunctional coating on the electrode. In this current design each of the selected component has specific role for deriving the reactive and porous coating for tailoring ionic environment, gas bubble wettability and adhesion. The natural abundant HNC having inherent hollow nano-structure contributes to form porous coating, AEPTMS acts as binder and residual acrylates from 5Acl makes the coating chemically reactive for post modifying with selected modifiers through 1,4-conjugate addition reaction. Post-modification with small hydrophilic and hydrophobic molecules tunes interfacial properties including gas bubble wettability, adhesion forces, zeta potential and surface free energy for improving overall water-splitting. The gradual increase in γ_{SG} improves hydration of modified coating and thereby γ_{SG} >48 mN m^{-1} ensures a capillary-force driven bubble detachment. Eventually, it improves gas removal from the electrode during electrolysis process. However, coating with highest γ_{SG} does not provide the best HER and OER performance, rather superaerophobic coating decorated with cationic and anionic moieties promotes HER and OER by providing their favorable microenvironments, and improved gas production by >60%. Thus, for coating with γ_{SG} >48 mN m^{-1}, the reaction microenvironment plays a pivotal role in governing the kinetics of both HER and OER. This is corroborated by the reduction in Tafel slopes for the respective reactions, indicating improved charge transfer kinetics. For a two-electrode based water-splitting set-up, the complementary uses of such superaerophobic coatings having different ionic environments as cathode and anodes, exhibits a synergistic effect, reducing the required potential to achieve current density of 100 mA cm^{-2} by nearly 500 ± 6 mV. Moreover, coatings with high γ_{SG} demonstrate robust underwater superoleophobicity, enabling stable electrolysis even in oil-contaminated electrolytes with negligible performance loss. Systematic introduction of molecular modifiers with varying charge density and dipolar character and pairing with charge-sensitive catalysts — including single-atom catalysts, heteroatom-doped materials may allow further tailoring of local electrostatic fields, interfacial pH, and ion-specific interactions is likely to provide a synergistic improvement in electrochemical water splitting.

4. Experimental Section

HNC (0.6 g) was dispersed in 20 mL of toluene, followed by the addition of 0.8 mL of AEPTMS under continuous stirring. The mixture was then heated to 60 °C with constant stirring for 2 h. After 2 h, the reaction was allowed to proceed at room temperature with continued stirring overnight. After 12 h, 3 mL of a previously prepared 5Acl solution—comprising 5.3 g of 5Acl dissolved in 40 mL of toluene—was added to the reaction mixture. After 2 h of further stirring, AEPTMS-functionalized nickel foam was immersed in the reaction mixture and kept for 30 min. The coated foams were then removed, air-dried under a compressed airflow, and subsequently cured at 70 °C overnight.

For surface free energy measurements, smooth surfaces were prepared as follows: a solution of AEPTMS (0.8 mL) was added to 20 mL of toluene, followed by the addition of 3 mL of a 5Acl solution, previously prepared by dissolving 5.3 g of 5Acl in 40 mL of toluene. From this reaction mixture, 100 µL was taken and uniformly coated onto AEPTMS-functionalized glass slides by doctor blade coating method.

Post-Modification of the Reactive Coating with Different Modifiers: Chemically reactive coatings on NF and glass were orthogonally post-modified with desired amines and acrylates (hydrophilic and hydrophobic). The residual acrylate groups of the coating can be modified with different amines, for instance AETMAC, β-ala, Glu, hexyl amine, octyl amine, and octadecyl amine. In this purpose, coated substrates (NF/ glass) were dipped in these amine solutions individually, i.e., AETMAC (5 mg mL^{-1} in water), Glu (2.5 mg mL^{-1} in water), β-ala (5 mg mL^{-1} in water), HA (5 mg mL^{-1} in ethanol), OA (5 mg mL^{-1} in ethanol), ODA (5 mg mL^{-1} in ethanol) for 12 h. Then the coated substrates were taken out and washed thoroughly with water or ethanol. After the amine treatment was done, the substrates were subsequently dipped in respective acrylate solutions to react with the residual amine groups of the coating, such as Sul.Ac. (5 mg mL^{-1} in water), DMAEA (20 µL mL^{-1} in water), hexyl acrylate (5 mg mL^{-1} in ethanol), octyl acrylate (5 mg mL^{-1} in ethanol), octadecyl acrylate (5 mg mL^{-1} in ethanol) for 12 h. Then the substrates were taken out and washed throughly in ethanol.

Reactive and modified coatings were characterized with standard techniques. Water and gas-bubble contact angles were measured using KRUSS Drop Shape Analyzer DSA25, the morphology of surfaces was investigated with FESEM; and ATR-FTIR and XPS were applied to reveal residual reactivity and postchemical modifications. Digital photographs and movies were captured with a Nikon COOLPIX B700 digital camera. Electrochemical measurements were conducted using a CH Instruments 600E series electrochemical workstation.

Statistical Analysis: All the experiments were performed in triplicate. Results were presented as mean ± SD.

Supporting Information

Supporting Information is available from the Wiley Online Library or from the author.

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Conflict of Interest

The authors declare no conflict of interest.

Author Contributions

J.D. performed all the experiments with the help of V.P., D.S., and S.D., J.D. designed experiments and analyzed data. D.D. provided theoretical studies on this work. U.M. conceived the idea, supervised the work, wrote the manuscript, and all authors contributed to editing and reviewing the manuscript.

Data Availability Statement

The data that support the findings of this study are available in the supplementary material of this article.

Keywords

bubble adhesion, electrochemical water splitting, gas-bubble repellence, ionic microenvironment, surface free energy

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